MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCHE200 - ORGANIC CHEMISTRY - 2

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Part A

Multiple choice questions. Answer all questions. Each question carries 1 marks

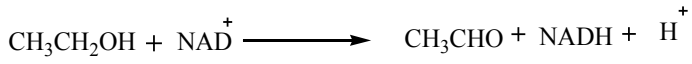
1. Electron-withdrawing substituents on the benzene ring:
   Increase acidity
2. Which of the following aldehydes will undergo an self-aldol condensation?
   Butanal
3. Explain how the inductive effect of an electron-withdrawing substituent affects carboxylic acid acidity.
   It stabilizes the conjugate base by delocalizing negative charge, increasing acidity
4. Ortho-nitrophenol is less soluble than para-nitrophenol due to:
   Intramolecular hydrogen bonding
5. Which of the following compounds has no α-hydrogen in aldehydes/ketones are acidic
   Benzaldehyde
6. The nitro groups in picric acid are present at:
   2,4,6-positions
7. What is the hybridization of carbonyl carbon changes after nucleophilic addition
   sp3
8. Identify the compound that is a dicarboxylic acid.
   Phthalic acid
9. Which of the following is an amide?
   CH_3CONH_2
10. Which alcohol will liberate hydrogen most rapidly with sodium?
    Methanol

Part B

Short Answer Questions. Answer any 4 questions. Each question carries 5 marks

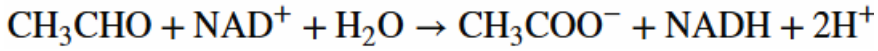
11. Explain the biochemical pathway of ethanol metabolism, naming the enzymes involved and the intermediate products formed. Why is acetaldehyde considered toxic?
    - In ethanol metabolism, human body breaks down and eliminates alcohol. The final products are carbon dioxide and water.

- The breakdown takes place in the liver.
- In the first step, ethanol is oxidized to acetaldehyde in the presence of the enzyme Alcohol Dehydrogenase



$$CH_3CH_2OH + NAD^+ \longrightarrow CH_3CHO + NADH + H^+$$

- Acetaldehyde is oxidized into acetate in the presence of the enzyme Aldehyde Dehydrogenase



$$CH_3CHO + NAD^+ + H_2O \rightarrow CH_3COO^- + NADH + 2H^+$$

- Acetic acid coenzyme A ligase (ACSL) breaks down acetic acid into acetyl-CoA, which is then broken down into carbon dioxide and water

Acetaldehyde is toxic because it rapidly binds to proteins and DNA, causes oxidative stress, and contributes to hangovers and liver damage

(Metabolism 4 marks+ toxicity of acetic acid 1 mark)

12. Write a note on the industrial uses of (a) formaldehyde (b) acetaldehyde

Formaldehyde

- *Manufacture of resins (phenol-formaldehyde, urea-formaldehyde)*
- *Used as disinfectant and preservative (formalin)*
- *Used in textile industry*
- *Used in manufacture of dyes, explosives, and rubber chemicals*

Acetaldehyde

- *Manufacture of acetic acid.*
- *Manufacture of perfumes and flavoring agents.*
- *Manufacture of pyridine and pyridine bases (pharmaceuticals).*
- *Manufacture of pentaerythritol (paints and explosives).*

(*2.5 marks + 2.5 marks*)

13. Sketch the curved arrow mechanism for the following conversions

(a) Ethyl acetate to Acetic acid (b) Acetic acid to Ethyl acetate

(a) Ethyl acetate → Acetic acid

CH COOC H + H O → CH COOH + C H OH (H / heat)

(b) Acetic acid → Ethyl acetate

CH COOH + C H OH → CH COOC H + H O (H / heat)

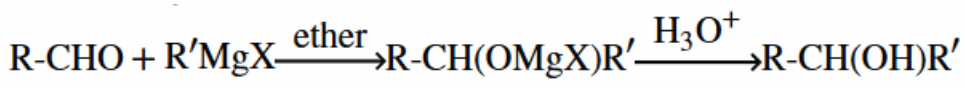
[2.5 mark – Correct chemical equation

2.5 mark – Basic mechanism (protonation + nucleophilic attack + elimination)

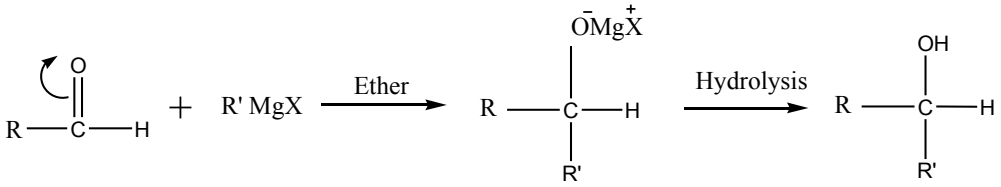
Even if full mechanism is not written, writing any two correct steps (protonation / nucleophilic attack) will get marks.]

14. How are secondary alcohols prepared using Grignard reagents? Explain the mechanism of reaction.

Secondary alcohols are synthesized by reacting a Grignard reagent RMgX with aldehydes except formaldehyde. The reaction involves the nucleophilic attack of the alkyl group on the aldehyde's carbonyl carbon, forming an alkoxide intermediate, which is then protonated with acid to yield the alcohol

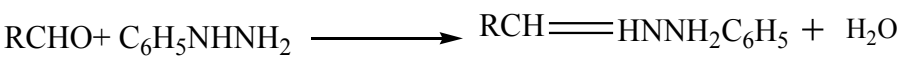


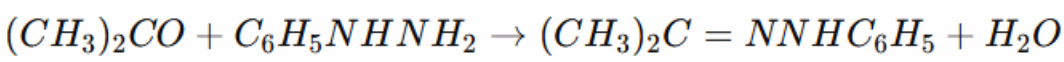
Mechanism



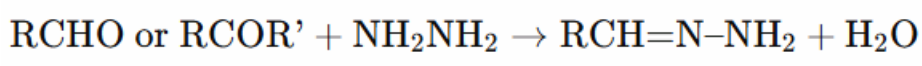
15. Write a note on reaction of phenyl hydrazine and hydrazine with ketone and aldehydes.
Reaction with Phenylhydrazine (C H NHNH)

Aldehydes and ketones react with phenyl hydrazine to form phenylhydrazones.

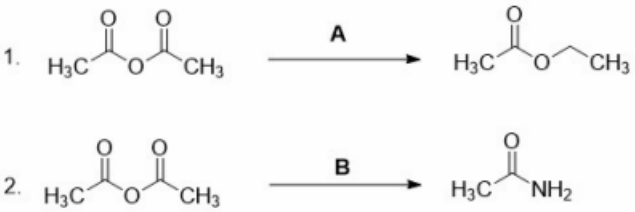




Aldehydes and ketones react with hydrazine to form **hydrazones**



16. Suggest suitable reagents. Sketch the curved arrow mechanism.



(1) Reaction A: Acetic anhydride → Ethyl acetate

Reagent: Ethanol (C H OH)

Equation:
(CH CO) O + C H OH → CH COOC H + CH COOH

(2) Reaction B: Acetic anhydride → Acetamide

Reagent: Ammonia (NH)

Equation:
(CH CO) O + NH → CH CONH + CH COOH

(Correct equation → 3 marks, Mechanism (any basic representation) → 2 marks)

Part C

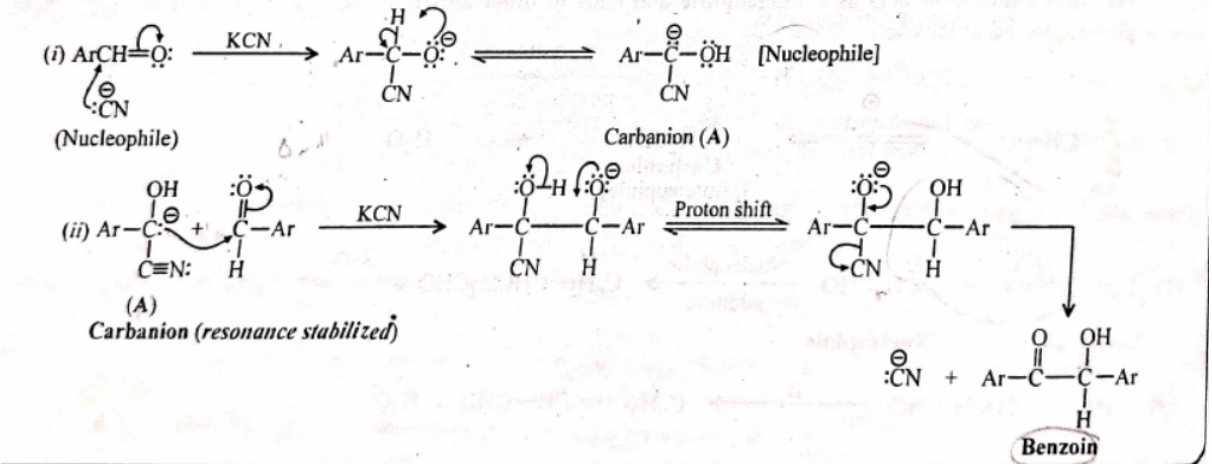
Short Essay Questions Answer any 2 questions Each question carries 10 marks

17. Explain the orientation and reactivity of phenol in electrophilic substitution reactions.
    - Phenol (C H OH) is **highly reactive** towards electrophilic substitution reactions compared to benzene.
    - The –OH group is an **ortho-para directing group**.
    - Phenol readily undergoes halogenation, nitration, and sulphonation, without a catalyst.

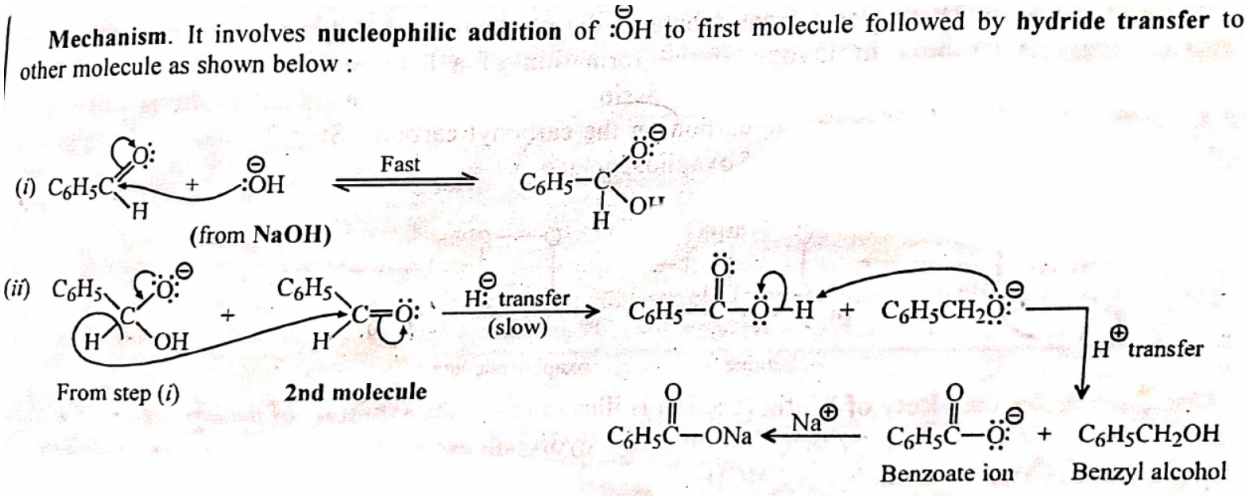
    (Correct explanation with mechanism 10 marks)
18. Draw the curved arrow mechanism of following reactions
    a) Benzoin condensation b) Cannizaro reaction. Discuss the salient features of the mechanisms.

    Benzoin condensation

Mechanism. It is believed that **nucleophilic addition** of cyanide ion to aromatic aldehyde generates a **carbanion** (a new ***nucleophile***) that adds to another molecule of aryl aldehyde followed by proton shift and removal of cyanide ion to form benzoin.



Cannizaro reaction



(Correct mechanism 5 marks each)

19. **Illustrate the detailed curved-arrow mechanisms for: (i) the conversion of a carboxylic acid into an acid chloride, and (ii) the conversion of a carboxylic acid into an acid anhydride. Clearly highlight the key mechanistic features involved in each transformation.**

    (i) Carboxylic acid → Acid chloride

    Reagent: SOCl (Thionyl chloride)

    Equation:
    R–COOH + SOCl → R–COCl + SO + HCl

    (ii) Carboxylic acid → Acid anhydride

    Reagent: P O

    Equation:
    2 R–COOH → (RCO) O + H O

[(i) Acid chloride (5 Marks): Reagent (SOCl) → 2 marks ; Equation → 1 mark; Mechanism (any basic steps) → 2 marks]

(ii) Acid anhydride (5 Marks); Reagent (P O) → 2 marks ; Equation → 1 mark; Mechanism (any simple steps) → 2 marks]